Herding quantum cats



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Our cat in three numbers

0.978

Fidelity of our 12-qubit cat on the simulator: 15× closer to a perfect cat than the organizers' baseline (0.41 in the same scorecard run).

0.913

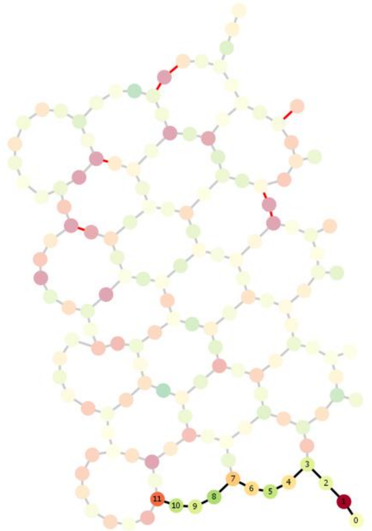
The same cat on the real ibm_kingston, against F = 0.599 for the baseline measured in the same job.

40

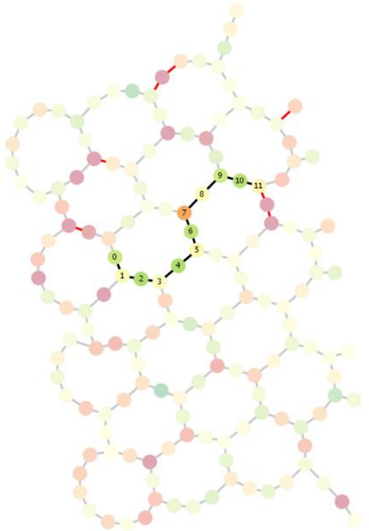
Qubits certified as genuinely entangled on real hardware (lower bound 0.56, above the 0.5 limit). We also certified 32.

Simulator: Fall Fest simulator of fake_kingston, 24,000 shots.
Hardware: ibm_kingston (IBM Heron r2), 10 Nov 2026.

Our recipe: four steps to a clean cat



Baseline: qubits 0–11



Ours: searched chain

1

Layout search. DFS with pruning over ~6,000 chains avoids the 16.8% readout qubit. F 0.44 → 0.72

2

Grow from the middle. Half the depth (11 → 6 CZ layers), idle time 5.9 → 2.4 µs. F → 0.74

3

TREX readout mitigation. Removes readout errors from P and X. F → 0.94

4

Linear ZNE (noise ×1, ×3). Extrapolates gate noise to zero. F → 0.978, R ≈ 15

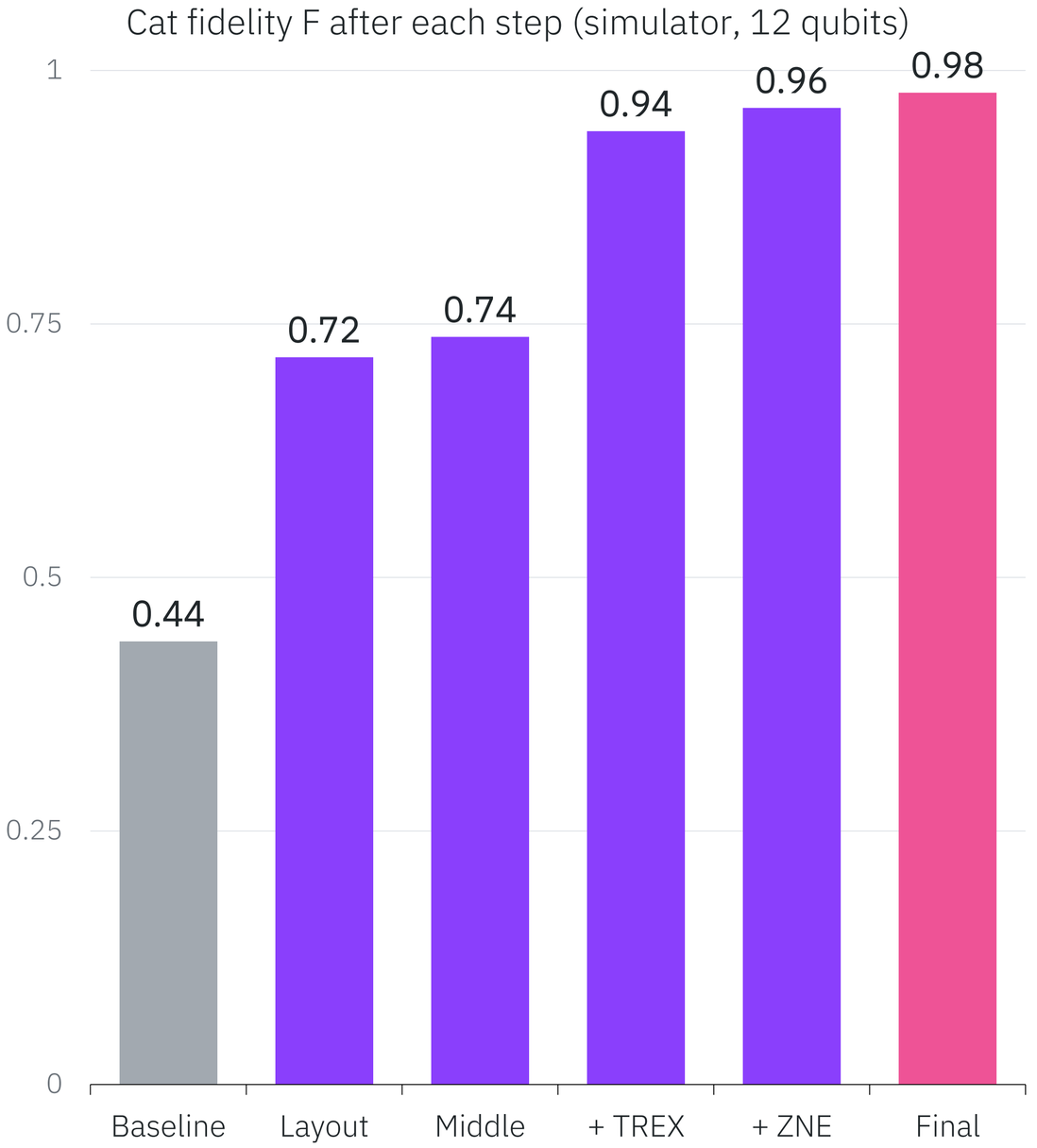
What helped, and by how much

Each step on the simulator, 12 qubits (Missions 2–6). F is a certified lower bound; above 0.5 means real entanglement.

Biggest single gain: TREX, error ε* 0.267 → 0.069, at the cost of ~2× larger error bars.

Tried and rejected: exponential ZNE (great once, unstable across seeds), gate twirling and DD (no gain on the simulator).

Final scorecard: ≈ 25/25 points, stable over 4 simulator seeds.



ibm_kingston · IBM Heron r2

On real hardware

Cleanest cat: 0.913

Our 12-qubit cat reached 0.913 ± 0.016, against 0.599 for the baseline in the same job (R = 3.9).

Calibration drift

Today the baseline qubits were good: F = 0.60 on hardware vs 0.41 on the simulator, so R fell from 15 to 3.9.

Biggest cat: 40 ✓

48 qubits: F = 0.435 ✗
40 qubits, first try: 0.405 ✗
40 qubits: F = 0.599 ± 0.018 ✓
32 qubits: F = 0.587 ± 0.013 ✓

Simulator vs reality

Crosstalk and coherent errors are missing from the simulator: our 48-qubit tree fell from 0.69 to 0.44.



40 qubits, entangled on a real chip



What we learned

Lessons

Qubit choice matters as much as the circuit.

Judge F together with its error bar.

Re-optimize for today's calibration.

Next

Error detection for bigger cats.

Thank you

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Questions?